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Augmenting the Crew: AI and Machine Learning in Fire and Emergency Services — A Research Agenda

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Abstract

Artificial intelligence and machine learning are reshaping emergency services, yet the dominant focus of both research and deployment has been on fire *physics* — spread prediction, detection, structural risk — and operational *logistics* — dispatch, routing, resource allocation. The layer that has received almost no systematic attention is the human team: how crews communicate, coordinate, and make decisions under acute stress, and how AI could augment rather than replace those processes. We present a structured landscape overview of AI/ML applications in fire and emergency services globally, organised by a two-axis typology (incident phase × unit of augmentation) that makes the gap visible. We describe a planned database of fire and emergency service organisations worldwide — seeded with a Netherlands pilot — built through web corpus curation (Hyphe), bibliographic harvesting (OpenAlex), and structured enrichment. On the basis of this landscape we propose seven research questions that constitute a research agenda for team-level AI augmentation in high-stakes emergency contexts, and we situate the agenda within the traditions of computer-supported cooperative work, naturalistic decision making, and crisis informatics.

Keywords: emergency services, fire services, artificial intelligence, machine learning, team augmentation, high-stakes collaboration, crisis informatics, research agenda, CSCW

1. Introduction

In the sixty years since Licklider's 1960 vision of man-computer symbiosis — the idea that human and machine should function as complementary partners, each contributing what the other cannot — the dominant realisation of that vision in safety-critical settings has been automation rather than augmentation. Autonomous systems replace humans in repetitive or dangerous tasks; decision-support systems present recommendations for human acceptance. What has been far less developed is the third possibility that Licklider envisioned: technology that amplifies the collective capability of *teams* — that makes crews smarter together than they are separately, that surfaces information at the right moment for the right person, that helps a team maintain shared awareness when stress, noise, and time pressure conspire against it.

Fire and emergency services are among the highest-stakes collaborative environments that exist. An incident commander managing a GRIP-1 structural fire is simultaneously tracking team positions, monitoring radio channels, updating a mental model of the fire's behaviour, and making allocation decisions under time pressure with imperfect information (Flin et al., 1996; Klein, 1998). Their team is distributed across multiple communication channels, operating in degraded sensory conditions, and subject to the cognitive and physiological effects of acute stress (Starcke & Brand, 2012). The quality of team coordination in this environment is not incidental to outcomes — it *is* the outcome, in the sense that most preventable adverse events in emergency response trace to coordination failure rather than individual error (Salas et al., 2005).

Against this background, the research literature on AI/ML in fire and emergency services presents a striking asymmetry. A substantial and growing body of work addresses fire detection from sensors and cameras (Muhammad et al., 2018; Yuan, 2020), fire spread prediction from weather and terrain models (Jolly et al., 2015), structural fire risk from building databases (Thomas & Corcoran, 2015), and dispatch and routing optimisation from historical incident logs (Kolesar & Walker, 1974). This work is valuable. But it addresses the fire as a physical and logistical problem. The team — the people who respond, who communicate, who make decisions, who debrief and learn — appears in this literature primarily as the recipient of better information, not as the locus of augmentation.

This paper argues that team-level augmentation is the most consequential and least-developed frontier in AI/ML for emergency services. We make three contributions. First, we present a structured landscape overview of existing AI/ML applications in fire and emergency services, organised by a typology that places team-level work on the same axes as individual-level and system-level work, making the gap visible. Second, we describe the design and initial implementation of a curated global database of fire and emergency service organisations and their AI/technology initiatives, beginning with the Netherlands and expanding outward. Third, on the basis of this landscape we propose seven research questions that constitute a concrete agenda for the next decade of team-augmentation research in emergency contexts.

We situate this agenda within three research traditions that have not been sufficiently integrated: computer-supported cooperative work (CSCW), which has developed the conceptual apparatus for understanding technology-mediated collaboration; naturalistic decision making (NDM), which has produced the richest accounts of expert cognition in high-stakes field settings; and crisis informatics, which has built the empirical and computational infrastructure for studying technology use during emergencies. Each tradition has something the others need. This paper is an argument for their convergence around the team in the fire.

2. Conceptual Framework

2.1 *Augmentation, automation, and the team*

The distinction between augmentation and automation is not merely terminological (Engelbart, 1962; Licklider, 1960). Automation removes a function from the human loop: the algorithm decides, the system acts. Augmentation keeps the human in the loop but changes what they can perceive, what they can process, and what they can do: the algorithm informs, the human decides. For safety-critical teams, the distinction matters enormously, for two reasons.

First, automation in high-stakes environments tends to produce skill degradation and automation surprise — the failure mode where operators lose the understanding needed to respond when the automated system fails (Bainbridge, 1983). In aviation, two generations of increasing cockpit automation have produced both safety gains and new failure modes, most of them involving crews who could no longer manually fly the aircraft when automation disengaged (Parasuraman et al., 2000). Fire services operate in environments that are far less predictable than commercial aviation, and where manual expertise cannot be surrendered to automation without unacceptable risk.

Second, the team is the unit of performance in emergency response, not the individual. Individual decision support that gives one firefighter better information but disrupts the shared mental model of the team may degrade collective performance even as it improves individual awareness (Salas et al., 2005). Team augmentation must be designed at the level of the team.

2.2 *Naturalistic decision making in emergency contexts*

The NDM tradition (Klein, 1998; Lipshitz et al., 2001) documents how experts make decisions under field conditions: time pressure, uncertainty, high stakes, and dynamic, ill-structured situations. The central finding is that expert decision making does not resemble the textbook rational-choice model. Experts pattern-match to a situation type, generate a single option, mentally simulate it, and act — or, if the simulation flags a problem, generate a revised option. This Recognition-Primed Decision (RPD) model (Klein, 1993) has been validated across incident command, military operations, intensive care, and firefighting.

For AI augmentation, the NDM finding is consequential: systems that require explicit option comparison slow experts down and disrupt their natural decision process. Effective team-level augmentation must work *with* the RPD process, not against it — surfacing cues that feed pattern recognition, flagging anomalies that might trigger simulation failure, and presenting information in forms that match the expert's situation model.

2.3 *Shared situation awareness and team cognition*

Endsley's 1995 model of situation awareness — perception of elements in the environment, comprehension of their meaning, and projection of their future state — was originally developed for individual operators. Endsley and Jones and Salas et al. extended it to *shared* situation awareness: the degree to which team members hold compatible situation models, and the degree to which that compatibility is necessary for effective team performance.

In emergency command, shared situation awareness degrades rapidly under stress, noise, and communication overload. The moment when a team loses shared awareness — when the commander's mental model of team positioning diverges from actual positioning, or when a GRIP step is assumed complete by one member and flagged as open by another — is a primary locus of adverse events. AI systems that help teams maintain shared awareness, that surface divergence before it becomes consequential, represent the highest-value augmentation opportunity that the field has yet to address systematically.

2.4 *Crisis informatics and high-stakes communication*

The crisis informatics tradition (Imran et al., 2015; Palen & Liu, 2007) has studied how information flows, is processed, and is acted upon during emergencies — primarily from the perspective of social media and public communication. Less attention has